Us app: Voice & Video Call Cost

Approximate monthly running cost of the app's own WebRTC calls at 10,000+ users · prepared 27 Oct 2026 · USD, with INR at about ₹95 per $

1. Short answer (10,000 users, typical use)

$33	**$236**	**$11,318**
₹3,200 a month	₹22,500 a month	₹10.8 L a month
Your own WebRTC + relay server (recommended)	Your own WebRTC, relay on Railway	Agora (what the app used before)

With **10,000 users** calling a typical amount, the calls cost about **$33 a month (₹3,200)** with your own WebRTC and a small relay server. The same calls on Agora would cost about **$11,318 a month (₹10.8 L)**. Most calls go straight from phone to phone, which costs nothing; you only pay to relay the calls that can't connect directly.

2. Where the cost comes from

Part	What it does	Cost
Direct calls (about 80% of calls)	Audio and video go straight between the two phones. Nothing passes through your servers.	**Free**
Relayed calls (about 20%)	Some mobile and office networks block direct connections. A TURN relay server forwards these calls. You pay for the data it sends.	**Main cost**, grows with video minutes
Call setup and ringing	Your Railway server rings the partner and swaps the connection details. These are a few small messages per call.	About $5–30 a month extra on Railway
Call notifications	Firebase Cloud Messaging buzzes the phone for incoming calls.	**Free**

3. Assumptions

A couples app has 2 users per couple, so **10,000 users = 5,000 couples**. Usage is the big unknown, so the estimate uses three levels:

Level	Couples who call on a given day	Video minutes per day	Voice minutes per day
Light	25%	10	10
Typical	**40%**	**20**	**15**
Heavy	60%	45	30

Technical figure	Value used	Why
Calls that need the relay	20%	The usual range is 10–20%; 20% is on the safe side for Indian mobile networks
Video call data	About 1.2 Mbps each way	Phone video at 720p (what the app asks for)
Voice call data	About 40 kbps each way	Opus voice codec, which WebRTC uses
Relay data per relayed minute	Video: 18 MB · voice: 0.6 MB	The relay forwards both directions
Busiest hour	15% of a day's calls	Used to size the relay server

4. Monthly cost by number of users

All three columns cover the **same calls**. Each figure is per month.

Users	Use	Video + voice minutes / month	Relay data	Own WebRTC + relay server (recommended)	Own WebRTC, relay on Railway	Agora
10,000	Light	0.8 M min	1.4 TB	**$29** **₹2,800**	$85 ₹8,100	$3,695 ₹3.5 L
	Typical	2.1 M min	4.4 TB	**$33** **₹3,200**	$236 ₹22,500	$11,318 ₹10.8 L
	Heavy	6.8 M min	14.9 TB	**$138** **₹13,100**	$760 ₹72,200	$37,625 ₹35.7 L
50,000	Light	3.8 M min	7.0 TB	**$69** **₹6,500**	$374 ₹35,500	$18,635 ₹17.7 L
	Typical	10.5 M min	22.1 TB	**$220** **₹20,900**	$1,132 ₹1.1 L	$56,750 ₹53.9 L
	Heavy	33.8 M min	74.5 TB	**$712** **₹67,700**	$3,751 ₹3.6 L	$188,285 ₹1.79 Cr
100,000	Light	7.5 M min	13.9 TB	**$154** **₹14,600**	$738 ₹70,100	$37,310 ₹35.4 L
	Typical	21.0 M min	44.3 TB	**$441** **₹41,900**	$2,254 ₹2.1 L	$113,540 ₹1.08 Cr
	Heavy	67.5 M min	149.0 TB	**$1,424** **₹1.4 L**	$7,492 ₹7.1 L	$376,610 ₹3.58 Cr

Recommended column: relay server(s) at DigitalOcean Bangalore ($24 each, 4 TB of data included, $10 per extra TB), plus the extra Railway cost for call setup. Railway column: about $10 for the relay plus $0.05 per GB of data. Agora column: list prices of $3.99 per 1,000 HD video minutes and $0.99 per 1,000 voice minutes, counted per person (so a 10-minute call between two people is 20 minutes), with 10,000 free minutes each month.

5. Relay servers you'd need (recommended setup)

Users	Typical use: busiest-hour relay traffic	Servers	Heavy use: busiest-hour relay traffic	Servers
10,000	49 Mbps	1 × $24	166 Mbps	1 × $24
50,000	246 Mbps	1 × $24	828 Mbps	3 × $24
100,000	492 Mbps	2 × $24	1,656 Mbps	6 × $24

Each $24 server (2 vCPU, 4 GB RAM) is planned for about 300 Mbps of relay traffic. Start with one; add another when monitoring shows it getting busy.

6. Good to know

- **Up to about 10,000 users, calls cost less than ₹5,000 a month in total**, because most calls never touch a server.
- **Video is about 30 times more data than voice**, so it drives the cost. Lowering video to 480p would cut relay data roughly in half.
- **The relay share is the biggest unknown.** If 30% of calls need the relay instead of 20%, relay costs rise by about half. After launch, the server can log which calls were relayed, so you can re-check this with real numbers.
- **Railway is fine for the app's server, but expensive for the relay**, because of its $0.05 per GB data price. It's about 5 times more than a DigitalOcean relay, and has no UDP, so call quality is worse.

- **Not included here:** the rest of the backend (chat, photos, database), which runs on Railway either way. At 10,000 users that's roughly $40–80 a month (server, Postgres, bucket storage), separate from calls.
- **Prices change.** These figures use list prices found on 27 Oct 2026 and an exchange rate of about ₹95 per $. They're estimates for planning, not quotes.

Sources

Agora video calling pricing: https://www.agora.io/en/pricing/agora-rtc
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